

Integration of a High-Throughput Digital Entropy Engine (MKRAND) into Quantum Control Systems

Technical Note for Zurich Instruments

November 23, 2025

Abstract

We present MKRAND, a deterministic digital circuit capable of producing 128 bits of perfectly uniform entropy every 256 clock cycles, with support for multiple independent or correlated output streams. This technical note outlines how integrating MKRAND into the Zurich Instruments Quantum Computing Control System (QCCS)—particularly at the FPGA layer of the SHFSG, SHFQA, SHFQC, and HDAWG—can provide researchers with an exceptionally stable, high-quality entropy source for real-time randomized benchmarking, noise spectroscopy, and calibration.

The result is a substantial reduction in experimental overhead related to entropy stabilization, letting researchers focus attention on the genuinely quantum aspects of the experiment rather than compensating for classical control-layer imperfections.

1 Introduction

Modern quantum processors rely on a sophisticated classical control stack to generate high-fidelity gate operations. The Zurich Instruments QCCS platform provides industry-leading microwave generation, readout, and arbitrary waveform synthesis through the SHFSG, SHFQA, SHFQC, and HDAWG modules.

While the control-layer is deterministic by design, many advanced quantum experiments—including randomized benchmarking (RB), Pauli twirling, noise spectroscopy, cross-talk analysis, and system identification—now require high-rate entropy from the classical layer.

Currently, these entropy requirements are typically fulfilled via software PRNGs or host-side QRNG devices, both of which introduce:

- upload overheads,
- lack of precise cycle-level synchronization across channels,
- subtle correlations introduced by pseudo-randomness,
- complexity in coordinating multi-module randomization.

MKRAND directly addresses these limitations.

2 The MKRAND Entropy Engine

MKRAND is a compact digital circuit designed for FPGA integration. It generates a full-width 128-bit entropy block every 256 clock cycles. Above several hundred MHz clock rates, MKRAND

provides entropy bandwidth in the hundreds of megabits per second, suitable for per-shot and even per-cycle randomization within the QCCS.

Key properties:

- **High Throughput:** 128 bits per 256 cycles (≈ 0.5 bits/cycle).
- **Intrinsic Uniformity:** Perfectly flat bit-distribution at each output word.
- **Multi-Stream Architecture:** A single MKRAND core can drive multiple independent 8–32 bit buses with programmable statistical independence.
- **Programmable Correlation:** Users can impose structured correlations or common-mode entropy across selected channels.
- **Deterministic Replay:** Any entropy pattern can be captured and replayed for reproducibility or regression testing.

These capabilities map naturally onto the requirements of multi-qubit quantum control.

3 Integration into Zurich Instruments QCCS

3.1 FPGA-Level Integration

The most natural integration point is inside the FPGA fabric of QCCS modules: SHFSG and SHFQC for gate synthesis; SHFQA for readout modulation; and HDAWG for flux and coupler control.

MKRAND can be exposed as:

- an **AXI-Stream interface** delivering fixed-width entropy words,
- a **parallel bus matrix** feeding channel-specific sequencers,
- a **register-programmable entropy controller** accessible through LabOne Q APIs.

Cycle-level proximity between entropy and waveform synthesis allows:

- real-time randomization,
- on-the-fly generation of Clifford sequences,
- pulse-level noise injection,
- gate-by-gate variation of amplitude, phase, and timing.

3.2 Software Stack Integration

Through LabOne Q, MKRAND appears as a *hardware randomization backend*. Existing RB, QPT, tomography, and noise routines can use its entropy streams directly by requesting random words instead of uploading precomputed sequences.

This moves randomness *inside* the control hardware, enabling:

- zero-latency random Clifford selection,
- real-time selection of random Pauli frames,
- correlated cross-channel noise generation for crosstalk mapping,
- per-shot randomization of readout weights and phases.

4 Correlated Entropy Across Physical Space

One of MKRAND’s differentiating capabilities is its programmable multi-stream correlation model. A single entropy source can drive all qubit control channels with:

1. **Independent streams** for studying local noise effects;
2. **Partially correlated streams** to emulate and diagnose intermediate-range environmental noise;
3. **Perfectly correlated streams** to characterize global error modes.

This feature allows researchers to *map noise structures in space* by observing how qubit error rates respond to engineered correlation patterns.

This is a major simplification compared to existing entropy stabilization mechanisms in which:

- noise must be injected independently on each module,
- software processes must coordinate phase and timing relationships,
- hardware must be synchronized via external triggers.

MKRAND centralizes this into a single block by ensuring:

- deterministic, synchronous entropy delivery,
- cross-module reproducibility,
- trivial correlation configuration.

5 Reduction of Experimental Overhead

Entropy generation and stabilization represent an often-overlooked tax on quantum computing experiments. Researchers spend significant time:

- validating PRNG uniformity,
- ensuring reproducibility of random sequences across experimental runs,
- aligning randomization across multiple hardware channels,
- compensating for pseudo-random correlations,
- managing upload bandwidth for large batches of random pulses.

Integrating MKRAND into the control layer eliminates these issues by making entropy:

- intrinsic to the hardware,
- cycle-synchronous,
- deterministic when required,
- configurable in terms of independence or correlation.

This shifts researcher effort away from classical-control complications and towards genuine quantum performance challenges such as:

- gate error suppression,
- decoherence mitigation,
- device modeling,
- qubit-qubit coupling characterization,
- scaling multi-qubit architectures.

6 Enabling New Quantum Experiments

The availability of high-rate, multi-channel entropy opens the door to experiments that are impractical under classical PRNG constraints:

- high-rate randomized benchmarking with per-shot unique Clifford gates;
- real-time randomized compiling for error suppression;
- structured noise injection for multi-qubit noise spectroscopy;
- scanning correlation patterns to identify coherent vs. incoherent noise components;
- stress-testing of control electronics under artificial entropy loads.

Because MKRAND’s output distribution is exactly uniform, users gain a statistically pristine randomness source that remains stable across:

- temperature drift,
- clock changes,
- FPGA reconfiguration,
- firmware updates.

This property allows experimental results to be attributed more cleanly to quantum behavior rather than control-layer variability.

7 Impact on the Future of Quantum and Classical Control

Integrating MKRAND does not replace quantum randomness; instead, it elevates the classical control layer to the point where it no longer constrains experiments that depend on statistical properties of random sequences.

This enables:

- clearer discrimination between classical and quantum noise sources,
- more precise modeling of non-Gaussian and correlated noise,
- improved estimation of spatial noise gradients,

- more efficient calibration cycles across large qubit arrays.

On the classical side, MKRAND offers a path for QCCS hardware to serve as a high-grade entropy provider for cryptographic demos, QRNG validation, and classically randomized protocols that support quantum algorithms.

8 Conclusion

MKRAND offers significant advantages to the Zurich Instruments QCCS platform by providing:

- a high-throughput, uniform entropy source,
- programmable multi-channel correlation,
- deterministic replay,
- FPGA-level timing precision.

These features simplify entropy-heavy calibration routines, accelerate randomized benchmarking, and enable new noise spectroscopy modalities. By stabilizing and simplifying the classical layer, MKRAND allows researchers to focus attention on the central challenges of scalable quantum computation.

Further Reading and Technical Resources

This technical note is accompanied by supporting whitepapers, technical specifications, and implementation examples. All materials are freely available at:

https://github.com/taguniversal/digital_blockchain_patents